机器人学
第六讲 雅可比矩阵与静力变换

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机器人逆向运动学

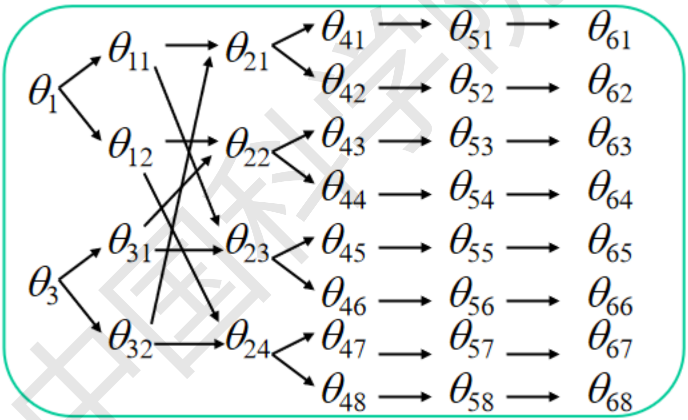
- 上次课内容提要
  - ➢球面坐标串联机器人的逆向运动学
  - ➢柱面坐标串联机器人的逆向运动学
  - ➢直角坐标串联机器人的逆向运动学
  - ➢并联机器人的逆向运动学
  - ➢冗余自由度机器人的逆向运动学
  - ➢机器人的微分运动学

1 球面坐标串联机器人的逆向运动学

➢ 球面坐标串联机器人的逆向运动学

1.1 解析法求解PUMA560机器人逆向运动学，最多8组解

θ_1、$\theta_3 \rightarrow \theta_2 \rightarrow \theta_4 \rightarrow \theta_5 \rightarrow \theta_6$



1.2 投影法与解析法结合求解Yaskawa机器人逆向运动学

投影法求解θ_1、θ_2、θ_3，先求解θ_1，再求解 θ_2和θ_3

解析法求解θ_4、θ_5、θ_6

2 柱面坐标串联机器人的逆向运动学

- ✓ p_z是平移关节的平移量
- ✓ p_x和p_y确定θ_1和θ_2的两组解
- ✓ $\theta_3=\theta-\theta_1-\theta_2$

3 直角坐标串联机器人的逆向运动学

- ✓ 利用由姿态矩阵转换为Z-Y-X欧拉角方法，求取旋转关节的关节角$\theta_4\sim\theta_6$
- ✓ 求解平移关节的位移量

4 并联机器人的逆向运动学

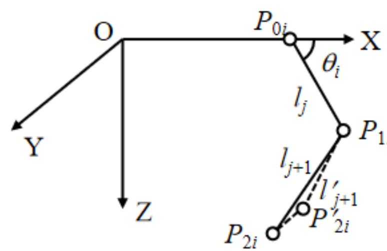
4.1 平移关节并联机器人的逆向运动学

- ✓ Stewart结构并联机器人：计算两点之间的距离
- ✓ 2-2-2正交结构并联机器人：

$$\begin{cases}\Delta l_1 = \Delta z - d\Delta\alpha \\ \Delta l_2 = \Delta z + d\Delta\alpha\end{cases},\begin{cases}\Delta l_3 = \Delta x - d\Delta\beta \\ \Delta l_4 = \Delta x + d\Delta\beta\end{cases},\begin{cases}\Delta l_5 = \Delta y - d\Delta\theta \\ \Delta l_6 = \Delta y + d\Delta\theta\end{cases}$$

4.2 旋转关节并联机器人的逆向运动学

- ✓ RRR平面并联机器人：每条运动支链是两连杆机器人
- ✓ Delta空间并联机器人：将P_{2i}投影到新的XOZ平面，每条运动支链是两连杆机器人



5 冗余自由度机器人的逆向运动学

5.1 自由度分配求解：移动操作手

5.2 约束求解：7自由度机械臂

6 机器人的微分运动

➢ 基坐标系下的微分变换

$$dT = \Delta T$$

$$\Delta = \begin{bmatrix} 0 & -\delta_z & \delta_y & d_x \\ \delta_z & 0 & -\delta_x & d_y \\ -\delta_y & \delta_x & 0 & d_z \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad 其中 \begin{cases} \delta_x = f_x d\theta \\ \delta_y = f_y d\theta \\ \delta_z = f_z d\theta \end{cases}$$

➢ 联体坐标系下的微分变换

$${}^T dT = T\,{}^T\Delta$$

$${}^T\Delta = \begin{bmatrix} 0 & -{}^T\delta_z & {}^T\delta_y & {}^Td_x \\ {}^T\delta_z & 0 & -{}^T\delta_x & {}^Td_y \\ -{}^T\delta_y & {}^T\delta_x & 0 & {}^Td_z \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad 其中 \begin{cases} {}^T\delta_x = {}^Tf_x\, {}^Td\theta \\ {}^T\delta_y = {}^Tf_y\, {}^Td\theta \\ {}^T\delta_z = {}^Tf_z\, {}^Td\theta \end{cases}$$

➢ 微分运动的等价变换

“基坐标系下的微分变换”转换为“联体坐标系下的微分变换”

$${}^T\Delta = T^{-1}\Delta T$$

$$= \begin{bmatrix} 0 & -\delta\cdot a & \delta\cdot o & \delta\cdot(p\times n)+d\cdot n \\ \delta\cdot a & 0 & -\delta\cdot n & \delta\cdot(p\times o)+d\cdot o \\ -\delta\cdot o & \delta\cdot n & 0 & \delta\cdot(p\times a)+d\cdot a \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{cases} {}^Td_x = \delta\cdot(p\times n)+d\cdot n = n\cdot(\delta\times p+d) \\ {}^Td_y = \delta\cdot(p\times o)+d\cdot o = o\cdot(\delta\times p+d), \\ {}^Td_z = \delta\cdot(p\times a)+d\cdot a = a\cdot(\delta\times p+d) \end{cases}$$

$$\begin{cases} {}^T\delta_x = \delta\cdot n \\ {}^T\delta_y = \delta\cdot o \\ {}^T\delta_z = \delta\cdot a \end{cases}$$

雅可比矩阵与静力变换

- 本次课内容提要
  - 机器人的雅可比矩阵(Jacobian matrix)
  - 机器人的雅可比矩阵实例
    - V-80机器人的雅可比矩阵
    - PUMA 560机器人的雅可比矩阵
  - 静力转换

1 机器人的雅可比矩阵

- **雅可比矩阵：** 机械手的笛卡儿空间运动速度与关节空间运动速度之间的变换称之为雅可比矩阵。关节空间向笛卡儿空间速度的传动比。

设x为表示机械手末端位姿的广义位置矢量，

q为机械手的关节坐标矢量，n个关节则为n维矢量

$$x = x(q) \Rightarrow \dot{x} = \begin{bmatrix} \sum_{j=1}^{n} \frac{\partial x_1}{\partial q_j}\dot{q}_j & \cdots & \sum_{j=1}^{n} \frac{\partial x_6}{\partial q_j}\dot{q}_j \end{bmatrix}^T = J(q)\dot{q}$$

$J(q)$为$6 \times n$的矩阵

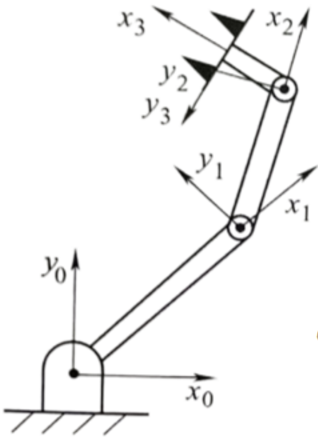
$$J_{ij}(q) = \frac{\partial x_i}{\partial q_j}$$

$$\dot{x} = \begin{bmatrix} v \\ w \end{bmatrix} = \lim_{\Delta t \to 0} \frac{1}{\Delta t}\begin{bmatrix} d \\ \delta \end{bmatrix} \Rightarrow D = \begin{bmatrix} d \\ \delta \end{bmatrix} = \lim_{\Delta t \to 0} \dot{x}\Delta t$$

$$D = \lim_{\Delta t \to 0} J(q)\dot{q}\Delta t = J(q)dq$$

$$\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} J_{11} & J_{12} & \vdots & J_{1n} \\ J_{21} & J_{22} & \vdots & J_{2n} \\ J_{31} & J_{32} & \vdots & J_{3n} \\ J_{41} & J_{42} & \vdots & J_{4n} \\ J_{51} & J_{52} & \vdots & J_{5n} \\ J_{61} & J_{62} & \vdots & J_{6n} \end{bmatrix} \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \vdots \\ \vdots \\ \dot{q}_{n-1} \\ \dot{q}_n \end{bmatrix}$$

1 机器人的雅可比矩阵



连杆坐标系

注：取末端坐标系O_3的Z轴和O_2的Z轴方向相同，O_3的X轴为夹指指向目标方向

D-H参数

i	θ_i	α_i	a_i	d_i
1	θ_1	0	l_1	0
2	θ_2	0	l_2	0
3	θ_3	0	l_3	0

$$ {}^{0}_{1}T = \begin{bmatrix} \cos\theta_1 & -\sin\theta_1 & 0 & l_1\cos\theta_1 \\ \sin\theta_1 & \cos\theta_1 & 0 & l_1\sin\theta_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad {}^{1}_{2}T = \begin{bmatrix} \cos\theta_2 & -\sin\theta_2 & 0 & l_2\cos\theta_2 \\ \sin\theta_2 & \cos\theta_2 & 0 & l_2\sin\theta_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^{2}_{3}T = \begin{bmatrix} \cos\theta_3 & -\sin\theta_3 & 0 & l_3\cos\theta_3 \\ \sin\theta_3 & \cos\theta_3 & 0 & l_3\sin\theta_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

1 机器人的雅可比矩阵

$$ {}^0_3T = {}^0_1T\,{}^1_2T\,{}^2_3T = \begin{bmatrix} cos(\theta_1+\theta_2+\theta_3) & -sin(\theta_1+\theta_2+\theta_3) & 0 & l_1cos\theta_1+l_2cos(\theta_1+\theta_2)+l_3cos(\theta_1+\theta_2+\theta_3) \\ sin(\theta_1+\theta_2+\theta_3) & cos(\theta_1+\theta_2+\theta_3) & 0 & l_1sin\theta_1+l_2sin(\theta_1+\theta_2)+l_3sin(\theta_1+\theta_2+\theta_3) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

基坐标下的机器人末端位姿表示

三轴坐标

$$ x = l_1cos\,\theta_1 + l_2cos(\,\theta_1+\theta_2) + l_3cos\,(\theta_1+\theta_2+\theta_3) $$

$$ y = l_1sin\,\theta_1 + l_2sin(\theta_1+\theta_2) + l_3sin(\theta_1+\theta_2+\theta_3) $$

$$ z = 0 $$

绕三轴转角 $\alpha = 0$ $\beta = 0$ $\gamma = \theta_1+\theta_2+\theta_3$

1 机器人的雅可比矩阵

$$x = x(q) \Rightarrow \dot{x} = \begin{bmatrix} \sum_{j=1}^{n} \frac{\partial x_1}{\partial q_j}\dot{q}_j & \cdots & \sum_{j=1}^{n} \frac{\partial x_6}{\partial q_j}\dot{q}_j \end{bmatrix}^T = J(q)\dot{q}$$

$J(q)$为$6 \times n$的矩阵

$$J_{ij}(q) = \frac{\partial x_i}{\partial q_j}$$

$$dx = \frac{\partial x}{\partial \theta_1} d\theta_1 + \frac{\partial x}{\partial \theta_2} d\theta_2 + \frac{\partial x}{\partial \theta_3} d\theta_1$$

$$= -[l_1 sin\theta_1 + l_2 sin(\theta_1+\theta_2) + l_3 sin(\theta_1+\theta_2+\theta_3)]d\theta_1 - [l_2 sin(\theta_1+\theta_2) + l_3 sin(\theta_1+\theta_2+\theta_3)]d\theta_2 - l_3 sin(\theta_1+\theta_2+\theta_3)d\theta_3$$

$$dy = \frac{\partial y}{\partial \theta_1} d\theta_1 + \frac{\partial y}{\partial \theta_2} d\theta_2 + \frac{\partial y}{\partial \theta_3} d\theta_3$$

$$= [l_1 cos\theta_1 + l_2 cos(\theta_1+\theta_2) + l_3 cos(\theta_1+\theta_2+\theta_3)]d\theta_1 + [l_2 cos(\theta_1+\theta_2) + l_3 cos(\theta_1+\theta_2+\theta_3)]d\theta_2 + l_3 cos(\theta_1+\theta_2+\theta_3)d\theta_3$$

$$d\gamma = \frac{\partial \gamma}{\partial \theta_1} d\theta_1 + \frac{\partial \gamma}{\partial \theta_2} d\theta_2 + \frac{\partial \gamma}{\partial \theta_3} d\theta_3$$

$$= d\theta_1 + d\theta_2 + d\theta_3$$

1 机器人的雅可比矩阵

将 dx、dy、$d\gamma$、$d\theta_1$、$d\theta_2$、$d\theta_3$ 写成时间导数，整理成如下矩阵

$$\begin{pmatrix}\dot{x}\\ \dot{y}\\ \dot{\gamma}\end{pmatrix}=J\begin{pmatrix}\dot{\theta}_1\\ \dot{\theta}_2\\ \dot{\theta}_3\end{pmatrix}=\begin{bmatrix}J_1 & J_2 & J_3\end{bmatrix}\begin{pmatrix}\dot{\theta}_1\\ \dot{\theta}_2\\ \dot{\theta}_3\end{pmatrix}=\begin{bmatrix}J_1\dot{\theta}_1 & J_2\dot{\theta}_2 & J_3\dot{\theta}_3\end{bmatrix}$$

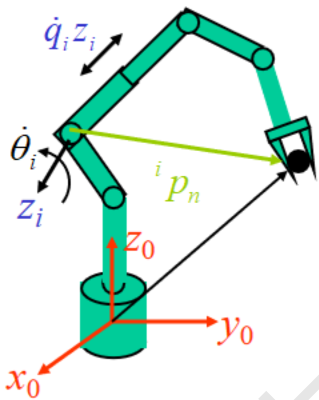
$$J=\begin{array}{c}\begin{bmatrix}-[l_1 sin\theta_1+l_2 sin(\theta_1+\theta_2)+l_3 sin(\theta_1+\theta_2+\theta_3)] & -[l_2 sin(\theta_1+\theta_2)+l_3 sin(\theta_1+\theta_2+\theta_3)] & -l_3 sin(\theta_1+\theta_2+\theta_3)\\ l_1 cos\theta_1+l_2 cos(\theta_1+\theta_2)+l_3 cos(\theta_1+\theta_2+\theta_3) & l_2 cos(\theta_1+\theta_2)+l_3 cos(\theta_1+\theta_2+\theta_3) & l_3 cos(\theta_1+\theta_2+\theta_3)\\ 1 & 1 & 1\end{bmatrix}\\ \begin{array}{ccc}\qquad\qquad J_1 \qquad\qquad & \qquad\qquad J_2 \qquad\qquad & \qquad J_3\end{array}\end{array}$$

其中 J 即为平面3R机器人相对基坐标系O的雅可比矩阵

1 机器人的雅可比矩阵

- 雅可比矩阵的求取

➢ 矢量积法　z_i是坐标系$\{i\}$的z轴在基坐标系$\{o\}$中的表示。



基坐标系

$$\begin{bmatrix} v \\ w \end{bmatrix} = \lim_{\Delta t \to 0} \frac{1}{\Delta t} \begin{bmatrix} d \\ \delta \end{bmatrix}$$

对于移动关节，有：$\delta = 0, d = z_i \dot{q} dt$

$$\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} z_i \\ 0 \end{bmatrix} \dot{q}_i, \quad J_i = \begin{bmatrix} z_i \\ 0 \end{bmatrix}$$

对于转动关节，有：$\delta = f d\theta = z_i \dot{\theta} dt = z_i \dot{q} dt, \ d = 0$

$^i p_n^0 = {}^0R_i \, {}^i p_n$　　$^i p_n^0$是$^i p_n$在基坐标系$\{o\}$中的表示。

$$\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} z_i \times {}^i p_n^0 \\ z_i \end{bmatrix} \dot{q}_i, \quad J_i = \begin{bmatrix} z_i \times {}^i p_n^0 \\ z_i \end{bmatrix} = \begin{bmatrix} z_i \times ({}^0R_i \, {}^i p_n) \\ z_i \end{bmatrix}$$

1 机器人的雅可比矩阵

➢ 矢量积法（续）

$$dT = T\Delta^T = \Delta T$$

z_i是坐标系$\{i\}$的z轴在基坐标系$\{o\}$中的表示：$z_i = f_x\bar{i} + f_y\bar{j} + f_z\bar{k}$

$$\begin{bmatrix} {}^0p_n \\ 1 \end{bmatrix} = \begin{bmatrix} {}^0R_i & {}^0p_i \\ 0 & 1 \end{bmatrix}\begin{bmatrix} {}^ip_n \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} \Delta & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} {}^0p_n \\ 1 \end{bmatrix} = \begin{bmatrix} \Delta & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} {}^0R_i & {}^0p_i \\ 0 & 1 \end{bmatrix}\begin{bmatrix} {}^ip_n \\ 1 \end{bmatrix}$$

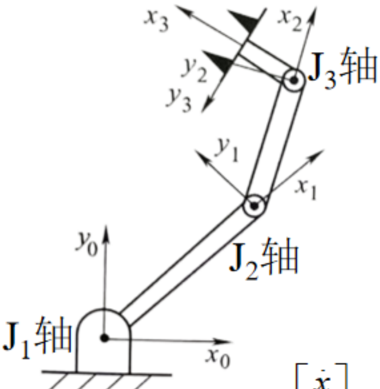
$$d\,{}^ip_n^0 = \Delta\,{}^0R_i\,{}^ip_n = {}^0R_i\,{}^i\Delta\,{}^ip_n$$

$$= \begin{bmatrix} 0 & -f_z d\theta & f_y d\theta \\ f_z d\theta & 0 & -f_x d\theta \\ -f_y d\theta & f_x d\theta & 0 \end{bmatrix}\begin{bmatrix} {}^ip_{nx}^0 \\ {}^ip_{ny}^0 \\ {}^ip_{nz}^0 \end{bmatrix} = \begin{bmatrix} -{}^ip_{ny}^0 f_z + {}^ip_{nz}^0 f_y \\ {}^ip_{nx}^0 f_z - {}^ip_{nz}^0 f_x \\ -{}^ip_{nx}^0 f_y + {}^ip_{ny}^0 f_x \end{bmatrix} = z_i \times {}^ip_n^0 d\theta$$

$$\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} z_i \times {}^ip_n^0 \\ z_i \end{bmatrix}\dot{q}_i, \quad J_i = \begin{bmatrix} z_i \times {}^ip_n^0 \\ z_i \end{bmatrix} = \begin{bmatrix} z_i \times ({}^0R_i\,{}^ip_n) \\ z_i \end{bmatrix}$$

$$\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} z_i \times {}^i p_n^0 \\ z_i \end{bmatrix} \dot{q}_i, \quad J_i = \begin{bmatrix} z_i \times {}^i p_n^0 \\ z_i \end{bmatrix} = \begin{bmatrix} z_i \times ({}^0 R_i \, {}^i p_n) \\ z_i \end{bmatrix}$$

对于J_1轴关节角 θ_1

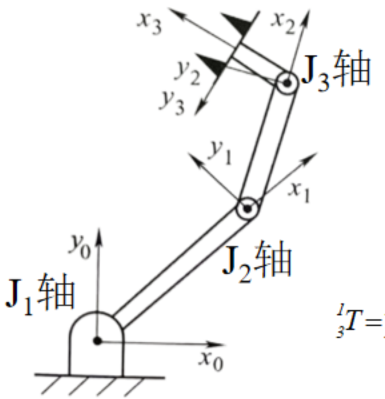


$$z_0 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \qquad {}^0 p_3^0 = \begin{bmatrix} l_1 \cos\theta_1 + l_2 \cos(\theta_1 + \theta_2) + l_3 \cos(\theta_1 + \theta_2 + \theta_3) \\ l_1 \sin\theta_1 + l_2 \sin(\theta_1 + \theta_2) + l_3 \sin(\theta_1 + \theta_2 + \theta_3) \\ 0 \end{bmatrix}$$

$$z_0 \times {}^0 p_3^0 = \begin{bmatrix} -[l_1 \sin\theta_1 + l_2 \sin(\theta_1 + \theta_2) + l_3 \sin(\theta_1 + \theta_2 + \theta_3)] \\ l_1 \cos\theta_1 + l_2 \cos(\theta_1 + \theta_2) + l_3 \cos(\theta_1 + \theta_2 + \theta_3) \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \\ \dot{\alpha} \\ \dot{\beta} \\ \dot{\gamma} \end{bmatrix} = J_1 \dot{\theta}_1 = \begin{bmatrix} -[l_1 \sin\theta_1 + l_2 \sin(\theta_1 + \theta_2) + l_3 \sin(\theta_1 + \theta_2 + \theta_3)] \\ l_1 \cos\theta_1 + l_2 \cos(\theta_1 + \theta_2) + l_3 \cos(\theta_1 + \theta_2 + \theta_3) \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \dot{\theta}_1$$

对于J$_2$轴关节角 θ_2



$$z_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \qquad {}^0_1T = \begin{bmatrix} \cos\theta_1 & -\sin\theta_1 & 0 & l_1\cos\theta_1 \\ \sin\theta_1 & \cos\theta_1 & 0 & l_1\sin\theta_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

${}^0R_1$ = upper-left 3×3 block of ${}^0_1T$

$${}^1_3T = {}^1_2T\,{}^2_3T = \begin{bmatrix} \cos(\theta_2+\theta_3) & -\sin(\theta_2+\theta_3) & 0 & l_3\cos(\theta_2+\theta_3)+l_2\cos\theta_2 \\ \sin(\theta_2+\theta_3) & \cos(\theta_2+\theta_3) & 0 & l_3\sin(\theta_2+\theta_3)+l_2\sin\theta_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

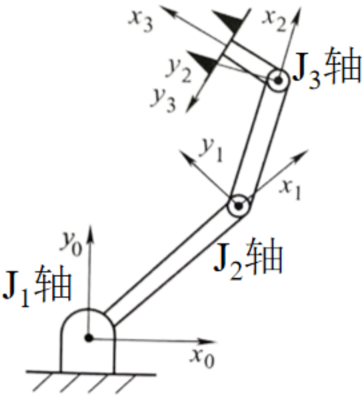
${}^1p_3$ = first three entries of the last column of ${}^1_3T$

$${}^1p_3^0 = {}^0R_1\,{}^1p_3 = \begin{bmatrix} l_3\cos(\theta_1+\theta_2+\theta_3)+l_2\cos(\theta_1+\theta_2) \\ l_3\sin(\theta_1+\theta_2+\theta_3)+l_2\sin(\theta_1+\theta_2) \\ 0 \end{bmatrix}$$

$$z_1\times{}^1p_3^0 = \begin{bmatrix} -[l_2\sin(\theta_1+\theta_2)+l_3\sin(\theta_1+\theta_2+\theta_3)] \\ l_2\cos(\theta_1+\theta_2)+l_3\cos(\theta_1+\theta_2+\theta_3) \\ 0 \end{bmatrix}$$

$$J_2 = \begin{bmatrix} z_1\times{}^1p_3^0 \\ z_1 \end{bmatrix}$$

对于J_3轴关节角 θ_3



$$
{}^0_2T = {}^0_1T\,{}^1_2T = \left[\begin{array}{ccc|c} \cos(\theta_1+\theta_2) & -\sin(\theta_1+\theta_2) & 0 & l_2\cos(\theta_1+\theta_2)+l_1\cos\theta_1 \\ \sin(\theta_1+\theta_2) & \cos(\theta_1+\theta_2) & 0 & l_2\sin(\theta_1+\theta_2)+l_1\sin\theta_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right]
$$

where the upper-left 3×3 block is ${}^0R_2$.

$$
z_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \qquad {}^2_3T = \left[\begin{array}{ccc|c} \cos\theta_3 & -\sin\theta_3 & 0 & l_3\cos\theta_3 \\ \sin\theta_3 & \cos\theta_3 & 0 & l_3\sin\theta_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right]
$$

where the boxed column is ${}^2p_3$.

$$
{}^2p_3^0 = {}^0R_2\,{}^2p_3 = \begin{bmatrix} l_3\cos(\theta_1+\theta_2+\theta_3) \\ l_3\sin(\theta_1+\theta_2+\theta_3) \\ 0 \end{bmatrix} \qquad z_2\times{}^2p_3^0 = \begin{bmatrix} -l_3\sin(\theta_1+\theta_2+\theta_3) \\ l_3\cos(\theta_1+\theta_2+\theta_3) \\ 0 \end{bmatrix}
$$

与求导方法获得的雅可比矩阵结果一致

$$
J_3 = \begin{bmatrix} z_2\times{}^2p_3^0 \\ z_2 \end{bmatrix} \qquad \begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} J_1 & J_2 & J_3 \end{bmatrix}\begin{bmatrix} \dot\theta_1 & \dot\theta_2 & \dot\theta_3 \end{bmatrix}^T
$$

1 机器人的雅可比矩阵

基坐标系下的雅可比矩阵$J(q)$和工具坐标系下的雅可比矩阵${}^{T}J(q)$的关系

$$\begin{bmatrix} v \\ w \end{bmatrix} = J(q)\dot{q}$$

$$v = {}_{n}^{0}R v^{n} \qquad v^{n} = {}_{n}^{0}R^{T} v$$

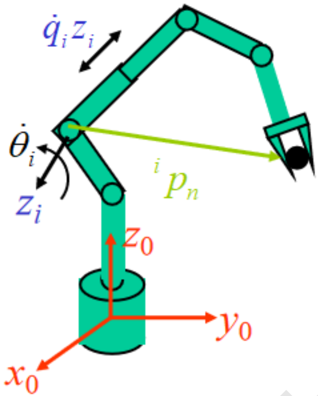
$$w = {}_{n}^{0}R w^{n} \qquad w^{n} = {}_{n}^{0}R^{T} w$$

$$\begin{bmatrix} v^{n} \\ w^{n} \end{bmatrix} = \begin{bmatrix} {}_{n}^{0}R^{T} & 0 \\ 0 & {}_{n}^{0}R^{T} \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} {}_{n}^{0}R^{T} & 0 \\ 0 & {}_{n}^{0}R^{T} \end{bmatrix} J(q)\dot{q} = {}^{T}J(q)\dot{q}$$

1 机器人的雅可比矩阵

➢ 微分变换法

$$\begin{cases} {}^T d_x = \delta\cdot(p\times n)+d\cdot n = n\cdot(\delta\times p+d) \\ {}^T d_y = \delta\cdot(p\times o)+d\cdot o = o\cdot(\delta\times p+d), \\ {}^T d_z = \delta\cdot(p\times a)+d\cdot a = a\cdot(\delta\times p+d) \end{cases} \quad \begin{cases} {}^T\delta_x = \delta\cdot n \\ {}^T\delta_y = \delta\cdot o \\ {}^T\delta_z = \delta\cdot a \end{cases}$$



工具坐标系

对于转动关节，有：

$$d=\begin{bmatrix}0\\0\\0\end{bmatrix},\delta=\begin{bmatrix}0\\0\\1\end{bmatrix}\mathrm{d}\theta_i$$

需要第i关节坐标系到末端坐标系的变换矩阵${}^iT_n$

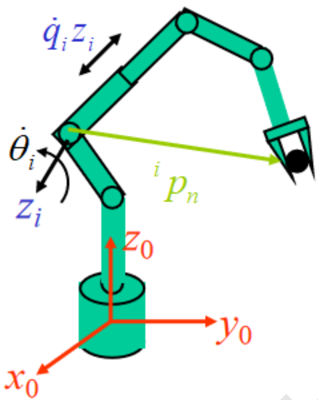
$$\begin{bmatrix} {}^T d_x \\ {}^T d_y \\ {}^T d_z \\ {}^T\delta_x \\ {}^T\delta_y \\ {}^T\delta_z \end{bmatrix} = \begin{bmatrix} (p\times n)_z \\ (p\times o)_z \\ (p\times a)_z \\ n_z \\ o_z \\ a_z \end{bmatrix}\mathrm{d}\theta_i, \qquad {}^T J_i = \begin{bmatrix} (p\times n)_z \\ (p\times o)_z \\ (p\times a)_z \\ n_z \\ o_z \\ a_z \end{bmatrix}$$

利用微分变换法求取关节的雅可比矩阵列时，将关节坐标系看作基坐标系

1 机器人的雅可比矩阵

➤ 微分变换法（续）

$$\begin{cases} {}^T d_x = \delta \cdot (p \times n) + d \cdot n = n \cdot (\delta \times p + d) \\ {}^T d_y = \delta \cdot (p \times o) + d \cdot o = o \cdot (\delta \times p + d), \\ {}^T d_z = \delta \cdot (p \times a) + d \cdot a = a \cdot (\delta \times p + d) \end{cases} \quad \begin{cases} {}^T \delta_x = \delta \cdot n \\ {}^T \delta_y = \delta \cdot o \\ {}^T \delta_z = \delta \cdot a \end{cases}$$



末端坐标系

对于移动关节，有：

$$d = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \mathrm{d}d_i, \delta = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

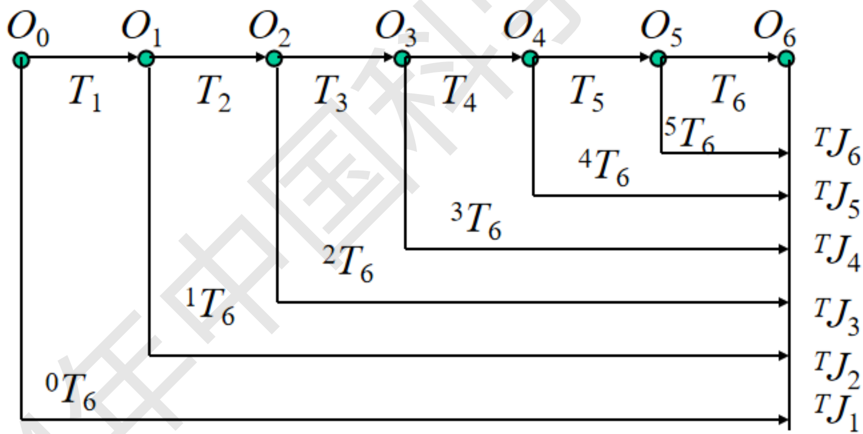
需要第i关节坐标系到末端坐标系的变换矩阵${}^i T_n$

$$\begin{bmatrix} {}^T d_x \\ {}^T d_y \\ {}^T d_z \\ {}^T \delta_x \\ {}^T \delta_y \\ {}^T \delta_z \end{bmatrix} = \begin{bmatrix} n_z \\ o_z \\ a_z \\ 0 \\ 0 \\ 0 \end{bmatrix} \mathrm{d}d_i, \qquad {}^T J_i = \begin{bmatrix} n_z \\ o_z \\ a_z \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

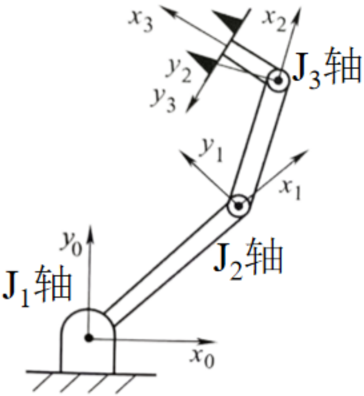
1 机器人的雅可比矩阵

➢ 微分变换法（续）

- ✓ 计算各个连杆间的变换矩阵 ${}^0T_1, {}^1T_2, \ldots, {}^{n-1}T_n$
- ✓ 计算各个连杆到末端连杆的变换矩阵 ${}^{n-1}T_n, {}^{n-2}T_n, \ldots, {}^0T_n$
- ✓ 计算${}^TJ(q)$的各列元素。根据关节i是移动关节或者转动关节，由${}^iT_n$计算${}^TJ_i$列。



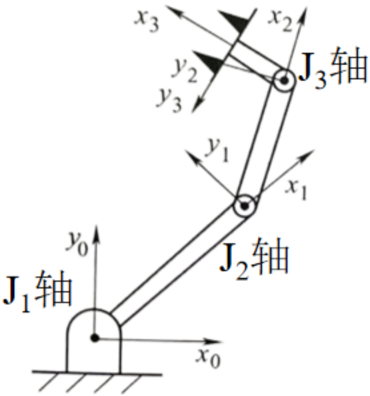
对于旋转关节J_3轴关节角 θ_3



$$
{}^2_3T = \begin{array}{c} \begin{array}{cccc} \mathbf{n} & \mathbf{o} & \mathbf{a} & \mathbf{p} \end{array} \\ \begin{bmatrix} \cos\theta_3 & -\sin\theta_3 & 0 & l_3\cos\theta_3 \\ \sin\theta_3 & \cos\theta_3 & 0 & l_3\sin\theta_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{array} \qquad d = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \delta = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} d\theta_3
$$

$$
{}^TJ_3 = \begin{bmatrix} {}^Td_x \\ {}^Td_y \\ {}^Td_z \\ {}^T\delta_x \\ {}^T\delta_y \\ {}^T\delta_z \end{bmatrix} = \begin{bmatrix} (p\times n)_z \\ (p\times o)_z \\ (p\times a)_z \\ n_z \\ o_z \\ a_z \end{bmatrix} d\theta_3 = \begin{bmatrix} 0 \\ l_3 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} d\theta_3
$$

对于旋转关节J_2轴关节角 θ_2



$$
{}^1_3T = {}^1_2T\,{}^2_3T = \begin{array}{c} \\ \\ \end{array}
\begin{array}{cccc}
\mathbf{n} & \mathbf{o} & \mathbf{a} & \mathbf{p} \\
\end{array}
$$

$$
{}^1_3T = {}^1_2T\,{}^2_3T = \begin{bmatrix} \cos(\theta_2+\theta_3) & -\sin(\theta_2+\theta_3) & 0 & l_3\cos(\theta_2+\theta_3)+l_2\cos\theta_2 \\ \sin(\theta_2+\theta_3) & \cos(\theta_2+\theta_3) & 0 & l_3\sin(\theta_2+\theta_3)+l_2\sin\theta_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$
d=\begin{bmatrix}0\\0\\0\end{bmatrix},\delta=\begin{bmatrix}0\\0\\1\end{bmatrix}d\theta_2
$$

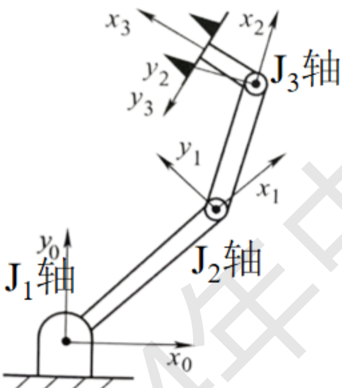
$$
{}^TJ_2 = \begin{bmatrix} {}^Td_x \\ {}^Td_y \\ {}^Td_z \\ {}^T\delta_x \\ {}^T\delta_y \\ {}^T\delta_z \end{bmatrix} = \begin{bmatrix} (p\times n)_z \\ (p\times o)_z \\ (p\times a)_z \\ n_z \\ o_z \\ a_z \end{bmatrix} d\theta_2 = \begin{bmatrix} l_2\sin\theta_3 \\ l_3+l_2\cos\theta_3 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} d\theta_2
$$

对于旋转关节J_1轴关节角 θ_1

$$
\begin{array}{cccc}
\mathbf{n} & \mathbf{o} & \mathbf{a} & \mathbf{p} \\
\end{array}
$$

$$
{}^{0}_{3}T=\begin{bmatrix}
cos(\theta_1+\theta_2+\theta_3) & -sin(\theta_1+\theta_2+\theta_3) & 0 & l_1cos\theta_1+l_2cos(\theta_1+\theta_2)+l_3cos(\theta_1+\theta_2+\theta_3) \\
sin(\theta_1+\theta_2+\theta_3) & cos(\theta_1+\theta_2+\theta_3) & 0 & l_1sin\theta_1+l_2sin(\theta_1+\theta_2)+l_3sin(\theta_1+\theta_2+\theta_3) \\
0 & 0 & 1 & 0 \\
0 & 0 & 0 & 1
\end{bmatrix}
$$

$$
d=\begin{bmatrix}0\\0\\0\end{bmatrix},\delta=\begin{bmatrix}0\\0\\1\end{bmatrix}d\theta_1
$$



$$
{}^{T}J_1=\begin{bmatrix}{}^{T}d_x\\{}^{T}d_y\\{}^{T}d_z\\{}^{T}\delta_x\\{}^{T}\delta_y\\{}^{T}\delta_z\end{bmatrix}=\begin{bmatrix}(p\times n)_z\\(p\times o)_z\\(p\times a)_z\\n_z\\o_z\\a_z\end{bmatrix}d\theta_1=\begin{bmatrix}l_1sin(\theta_2+\theta_3)+l_2sin\theta_3\\l_1cos(\theta_2+\theta_3)+l_2cos\theta_3+l_3\\0\\0\\0\\1\end{bmatrix}d\theta_1
$$

1 机器人的雅可比矩阵

$$\begin{bmatrix} {}^{T}d_x \\ {}^{T}d_y \\ {}^{T}d_z \\ {}^{T}\delta_x \\ {}^{T}\delta_y \\ {}^{T}\delta_z \end{bmatrix} = {}^{T}J(q)\dot{q} = \begin{bmatrix} 0 & l_2 sin\theta_3 & l_1 sin(\theta_2+\theta_3)+l_2 sin\theta_3 \\ l_3 & l_2 cos\theta_3 + l_3 & l_1 cos(\theta_2+\theta_3)+l_2 cos\theta_3 + l_3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} d\theta_1 \\ d\theta_2 \\ d\theta_3 \end{bmatrix}$$

1 机器人的雅可比矩阵

- 基于速度雅可比的机器人奇异性分析

$$\dot{X} = J(q)\dot{q} \qquad \dot{q} = J^{-1}(q)\dot{X}$$

$det[J(q)] \neq 0$ $J(q)$ 可逆

$det[J(q)] = 0$ $J(q)$ 不可逆，即奇异性，对应位形成为奇异位形

- 串联机器人在其工作空间的边界均为奇异位形，大多数机器人在工作空间内也存在奇异位形
- 奇异位形处，速度雅可比矩阵降秩，末端执行器失去一个或几个自由度（笛卡尔空间）
- 控制角度，处于奇异位形时，无论多大关节速度，都不能使机器人末端运动，串联机器人接近奇异位形时，速度会趋于无穷大

1 机器人的雅可比矩阵

$$\begin{pmatrix}\dot{x}\\ \dot{y}\\ \dot{\gamma}\end{pmatrix}=J\begin{pmatrix}\dot{\theta}_1\\ \dot{\theta}_2\\ \dot{\theta}_3\end{pmatrix}=\begin{bmatrix}J_1 & J_2 & J_3\end{bmatrix}\begin{pmatrix}\dot{\theta}_1\\ \dot{\theta}_2\\ \dot{\theta}_3\end{pmatrix}=\begin{bmatrix}J_1\dot{\theta}_1 & J_2\dot{\theta}_2 & J_3\dot{\theta}_3\end{bmatrix}$$

$$J=\begin{bmatrix}-[l_1 sin\theta_1+l_2 sin(\theta_1+\theta_2)+l_3 sin(\theta_1+\theta_2+\theta_3)] & -[l_2 sin(\theta_1+\theta_2)+l_3 sin(\theta_1+\theta_2+\theta_3)] & -l_3 sin(\theta_1+\theta_2+\theta_3)\\ l_1 cos\theta_1+l_2 cos(\theta_1+\theta_2)+l_3 cos(\theta_1+\theta_2+\theta_3) & l_2 cos(\theta_1+\theta_2)+l_3 cos(\theta_1+\theta_2+\theta_3) & l_3 cos(\theta_1+\theta_2+\theta_3)\\ 1 & 1 & 1\end{bmatrix}$$

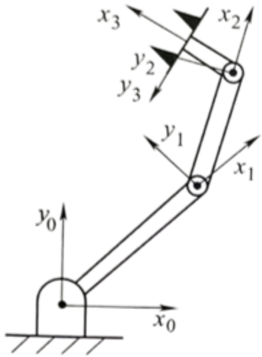
求解

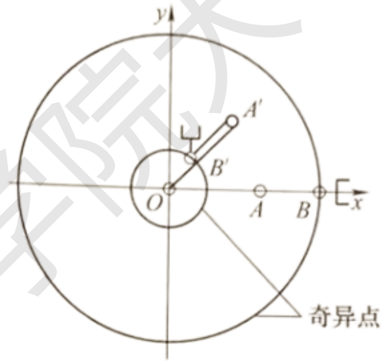
$$\det(J)=0$$

$$=\begin{vmatrix}-[\mathrm{l}_1\sin\theta_1+\mathrm{l}_2\sin(\theta_1+\theta_2)+\mathrm{l}_3\sin(\theta_1+\theta_2+\theta_3)] & -[\mathrm{l}_2\sin(\theta_1+\theta_2)+\mathrm{l}_3\sin(\theta_1+\theta_2+\theta_3)] & -\mathrm{l}_3\sin(\theta_1+\theta_2+\theta_3)\\ \mathrm{l}_1\cos\theta_1+\mathrm{l}_2\cos(\theta_1+\theta_2)+\mathrm{l}_3\cos(\theta_1+\theta_2+\theta_3) & \mathrm{l}_2\cos(\theta_1+\theta_2)+\mathrm{l}_3\cos(\theta_1+\theta_2+\theta_3) & \mathrm{l}_3\cos(\theta_1+\theta_2+\theta_3)\\ 1 & 1 & 1\end{vmatrix}$$

解得 $$l_1 l_2 sin\theta_2=0$$

1 机器人的雅可比矩阵





$\theta_2 = 0$

机器人失去一个自由度，机器人处于完全展开状态

$\theta_2 = \pi$

机器人失去一个自由度，机器人处于折叠状态

2 机器人的雅可比矩阵实例

2.1 V80机器人的雅可比矩阵

连杆	关节角 θ	扭转角 α	连杆长度 a	偏移量 d
1	θ_1	-90°	0	0
2	θ_2	0°	a_2	0
3	θ_3	90°	a_3	0
4	θ_4	-90°	0	d_4
5	θ_5	90°	0	0
6	θ_6	0°	0	0

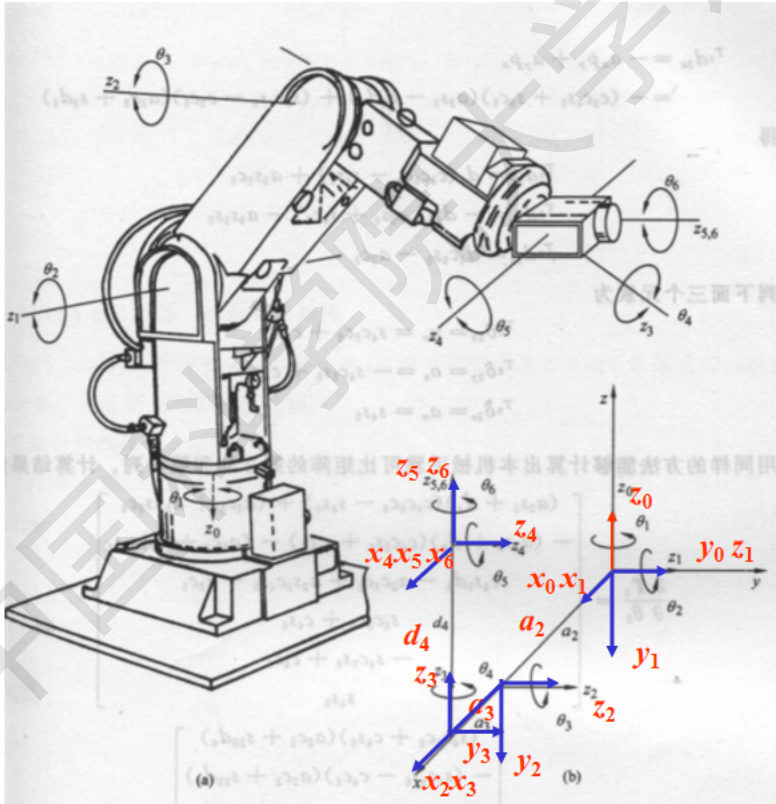


图 3.15 V-80 机械手外形图及停止位置

2.1 V80机器人的雅可比矩阵

- V-80机器人的雅可比矩阵
  - ✓ 坐标系与连杆参数

连杆	关节角θ	扭转角α	连杆长度a	偏移量d
1	θ_1	-90°	0	0
2	θ_2	0°	a_2	0
3	θ_3	90°	a_3	0
4	θ_4	-90°	0	d_4
5	θ_5	90°	0	0
6	θ_6	0°	0	0
	绕z轴	绕x轴	沿x轴	沿z轴

2.1 V80机器人的雅可比矩阵

➤ V-80机器人的雅可比矩阵

✓ 连杆变换矩阵

$$T_i = \begin{bmatrix} \cos\theta_i & -\sin\theta_i\cos\alpha_i & \sin\theta_i\sin\alpha_i & a_i\cos\theta_i \\ \sin\theta_i & \cos\theta_i\cos\alpha_i & -\cos\theta_i\sin\alpha_i & a_i\sin\theta_i \\ 0 & \sin\alpha_i & \cos\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_1 = \begin{bmatrix} \cos\theta_1 & 0 & -\sin\theta_1 & 0 \\ \sin\theta_1 & 0 & \cos\theta_1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},\ T_2 = \begin{bmatrix} \cos\theta_2 & -\sin\theta_2 & 0 & a_2\cos\theta_2 \\ \sin\theta_2 & \cos\theta_2 & 0 & a_2\sin\theta_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},\ T_3 = \begin{bmatrix} \cos\theta_3 & 0 & \sin\theta_3 & a_3\cos\theta_3 \\ \sin\theta_3 & 0 & -\cos\theta_3 & a_3\sin\theta_3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_4 = \begin{bmatrix} \cos\theta_4 & 0 & -\sin\theta_4 & 0 \\ \sin\theta_4 & 0 & \cos\theta_4 & 0 \\ 0 & -1 & 0 & d_4 \\ 0 & 0 & 0 & 1 \end{bmatrix},\quad T_5 = \begin{bmatrix} \cos\theta_5 & 0 & \sin\theta_5 & 0 \\ \sin\theta_5 & 0 & -\cos\theta_5 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},\ T_6 = \begin{bmatrix} \cos\theta_6 & -\sin\theta_6 & 0 & 0 \\ \sin\theta_6 & \cos\theta_6 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2.1 V80机器人的雅可比矩阵

✓ 连杆到机器人末端的变换矩阵

$$
{}^{5}T_6 = T_6 = \begin{bmatrix} \cos\theta_6 & -\sin\theta_6 & 0 & 0 \\ \sin\theta_6 & \cos\theta_6 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$
{}^{4}T_6 = T_5 T_6 = \begin{bmatrix} \cos\theta_5 & 0 & \sin\theta_5 & 0 \\ \sin\theta_5 & 0 & -\cos\theta_5 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\theta_6 & -\sin\theta_6 & 0 & 0 \\ \sin\theta_6 & \cos\theta_6 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos\theta_5\cos\theta_6 & -\cos\theta_5\sin\theta_6 & \sin\theta_5 & 0 \\ \sin\theta_5\cos\theta_6 & -\sin\theta_5\sin\theta_6 & -\cos\theta_5 & 0 \\ \sin\theta_6 & \cos\theta_6 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$
{}^{3}T_6 = T_4\,{}^{4}T_6 = \begin{bmatrix} \cos\theta_4 & 0 & -\sin\theta_4 & 0 \\ \sin\theta_4 & 0 & \cos\theta_4 & 0 \\ 0 & -1 & 0 & d_4 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\theta_5\cos\theta_6 & -\cos\theta_5\sin\theta_6 & \sin\theta_5 & 0 \\ \sin\theta_5\cos\theta_6 & -\sin\theta_5\sin\theta_6 & -\cos\theta_5 & 0 \\ \sin\theta_6 & \cos\theta_6 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$
= \begin{bmatrix} \cos\theta_4\cos\theta_5\cos\theta_6 - \sin\theta_4\sin\theta_6 & -\cos\theta_4\cos\theta_5\sin\theta_6 - \sin\theta_4\cos\theta_6 & \cos\theta_4\sin\theta_5 & 0 \\ \sin\theta_4\cos\theta_5\cos\theta_6 + \cos\theta_4\sin\theta_6 & -\sin\theta_4\cos\theta_5\sin\theta_6 + \cos\theta_4\cos\theta_6 & -\sin\theta_4\cos\theta_5 & 0 \\ -\sin\theta_5\cos\theta_6 & \sin\theta_5\sin\theta_6 & \cos\theta_5 & d_4 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

2.1 V80机器人的雅可比矩阵

✓Jacobian矩阵：以末端坐标系表示

$$d=\begin{bmatrix}0\\0\\0\end{bmatrix},\delta=\begin{bmatrix}0\\0\\1\end{bmatrix}\mathrm{d}\theta_i$$

$$J_i=\begin{bmatrix}(p\times n)_z\\(p\times o)_z\\(p\times a)_z\\n_z\\o_z\\a_z\end{bmatrix}\quad J_6=\begin{bmatrix}0\\0\\0\\0\\0\\1\end{bmatrix}\quad J_5=\begin{bmatrix}0\\0\\0\\\sin\theta_6\\\cos\theta_6\\0\end{bmatrix}\quad p_4=\begin{bmatrix}0\\0\\d_4\end{bmatrix},\ \begin{bmatrix}(p_4\times n)_z\\(p_4\times o)_z\\(p_4\times a)_z\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix}\Rightarrow J_4=\begin{bmatrix}0\\0\\0\\-\sin\theta_5\cos\theta_6\\\sin\theta_5\sin\theta_6\\\cos\theta_5\end{bmatrix}$$

$$J_3=\begin{bmatrix}d_4(\cos\theta_4\cos\theta_5\cos\theta_6-\sin\theta_4\sin\theta_6)+a_3\sin\theta_5\cos\theta_6\\-d_4(\cos\theta_4\cos\theta_5\sin\theta_6+\sin\theta_4\cos\theta_6)-a_3\sin\theta_5\cos\theta_6\\d_4\cos\theta_4\sin\theta_5-a_3\cos\theta_5\\\sin\theta_4\cos\theta_5\cos\theta_6+\cos\theta_4\sin\theta_6\\-(\sin\theta_4\cos\theta_5\sin\theta_6-\cos\theta_4\cos\theta_6)\\\sin\theta_4\sin\theta_5\end{bmatrix}$$

2.1 V80机器人的雅可比矩阵

✓Jacobian矩阵

$$
J_2 = \begin{bmatrix}
(a_2 \sin\theta_3 + d_4)(\cos\theta_4 \cos\theta_5 \cos\theta_6 - \sin\theta_4 \sin\theta_6) + (a_2 \cos\theta_3 + a_3)\sin\theta_5 \cos\theta_6 \\
-(a_2 \sin\theta_3 + d_4)(\cos\theta_4 \cos\theta_5 \sin\theta_6 + \sin\theta_4 \cos\theta_6) - (a_2 \cos\theta_3 + a_3)\sin\theta_5 \cos\theta_6 \\
d_4 \cos\theta_4 \sin\theta_5 - a_2 \cos\theta_3 \cos\theta_5 + a_2 \sin\theta_3 \cos\theta_4 \sin\theta_5 - a_3 \cos\theta_5 \\
\sin\theta_4 \cos\theta_5 \cos\theta_6 + \cos\theta_4 \sin\theta_6 \\
-(\sin\theta_4 \cos\theta_5 \sin\theta_6 - \cos\theta_4 \cos\theta_6) \\
\sin\theta_4 \sin\theta_5
\end{bmatrix}
$$

$$
J_1 = \begin{bmatrix}
(\sin\theta_4 \cos\theta_5 \cos\theta_6 + \cos\theta_4 \sin\theta_6)[a_2 \cos\theta_2 + d_4 \sin(\theta_2 + \theta_3)] \\
-(\sin\theta_4 \cos\theta_5 \sin\theta_6 - \cos\theta_4 \cos\theta_6)[a_2 \cos\theta_2 + d_4 \sin(\theta_2 + \theta_3)] \\
\sin\theta_4 \sin\theta_5 [a_2 \cos\theta_2 + d_4 \sin(\theta_2 + \theta_3)] \\
-\sin(\theta_2 + \theta_3)(\cos\theta_4 \cos\theta_5 \cos\theta_6 - \sin\theta_4 \sin\theta_6) + \cos(\theta_2 + \theta_3)\sin\theta_5 \sin\theta_6 \\
\sin(\theta_2 + \theta_3)(\cos\theta_4 \cos\theta_5 \sin\theta_6 + \sin\theta_4 \cos\theta_6) - \cos(\theta_2 + \theta_3)\sin\theta_5 \cos\theta_6 \\
-\sin(\theta_2 + \theta_3)\cos\theta_4 \sin\theta_5 + \cos(\theta_2 + \theta_3)\cos\theta_5
\end{bmatrix}
$$

2.2 PUMA560机器人的雅可比矩阵

➢ PUMA 560 机器人的雅可比矩

$$ {}^5T_6 = T_6 = \begin{bmatrix} \cos\theta_6 & -\sin\theta_6 & 0 & 0 \\ \sin\theta_6 & \cos\theta_6 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad J_6 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}^T $$

$$ {}^4T_6 = T_5 T_6 = \begin{bmatrix} \cos\theta_5\cos\theta_6 & -\cos\theta_5\sin\theta_6 & -\sin\theta_5 & 0 \\ \sin\theta_5\cos\theta_6 & -\sin\theta_5\sin\theta_6 & \cos\theta_5 & 0 \\ -\sin\theta_6 & -\cos\theta_6 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad J_5 = \begin{bmatrix} 0 & 0 & 0 & -\sin\theta_6 & -\cos\theta_6 & 0 \end{bmatrix}^T $$

$$ {}^3T_6 = T_4 T_5 T_6 $$

$$ = \begin{bmatrix} \cos\theta_4\cos\theta_5\cos\theta_6 - \sin\theta_4\sin\theta_6 & -\cos\theta_4\cos\theta_5\sin\theta_6 - \sin\theta_4\cos\theta_6 & -\cos\theta_4\sin\theta_5 & 0 \\ \sin\theta_4\cos\theta_5\cos\theta_6 + \cos\theta_4\sin\theta_6 & -\sin\theta_4\cos\theta_5\sin\theta_6 + \cos\theta_4\cos\theta_6 & -\sin\theta_4\sin\theta_5 & 0 \\ \sin\theta_5\cos\theta_6 & -\sin\theta_5\sin\theta_6 & \cos\theta_5 & d_4 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ J_4 = \begin{bmatrix} 0 & 0 & 0 & \sin\theta_5\cos\theta_6 & -\sin\theta_5\sin\theta_6 & \cos\theta_5 \end{bmatrix}^T $$

2.2 PUMA560机器人的雅可比矩阵

$$^{2}T_6 = T_3 T_4 T_5 T_6$$

$$= \begin{bmatrix} \cos\theta_3 & 0 & -\sin\theta_3 & a_3\cos\theta_3 \\ \sin\theta_3 & 0 & \cos\theta_3 & a_3\sin\theta_3 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\theta_4\cos\theta_5\cos\theta_6 - \sin\theta_4\sin\theta_6 & -\cos\theta_4\cos\theta_5\sin\theta_6 - \sin\theta_4\cos\theta_6 & -\cos\theta_4\sin\theta_5 & 0 \\ \sin\theta_4\cos\theta_5\cos\theta_6 + \cos\theta_4\sin\theta_6 & -\sin\theta_4\cos\theta_5\sin\theta_6 + \cos\theta_4\cos\theta_6 & -\sin\theta_4\sin\theta_5 & 0 \\ \sin\theta_5\cos\theta_6 & -\sin\theta_5\sin\theta_6 & \cos\theta_5 & d_4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \begin{array}{l} \cos\theta_3(\cos\theta_4\cos\theta_5\cos\theta_6 - \sin\theta_4\sin\theta_6) \\ -\sin\theta_3\sin\theta_5\cos\theta_6 \end{array} & \begin{array}{l} -\cos\theta_3(\cos\theta_4\cos\theta_5\sin\theta_6 + \sin\theta_4\cos\theta_6) \\ +\sin\theta_3\sin\theta_5\sin\theta_6 \end{array} & \begin{array}{l} -\cos\theta_3\cos\theta_4\sin\theta_5 \\ -\sin\theta_3\cos\theta_5 \end{array} & \begin{array}{l} -d_4\sin\theta_3 \\ +a_3\cos\theta_3 \end{array} \\ \begin{array}{l} \sin\theta_3(\cos\theta_4\cos\theta_5\cos\theta_6 - \sin\theta_4\sin\theta_6) \\ +\cos\theta_3\sin\theta_5\cos\theta_6 \end{array} & \begin{array}{l} -\sin\theta_3(\cos\theta_4\cos\theta_5\sin\theta_6 + \sin\theta_4\cos\theta_6) \\ -\cos\theta_3\sin\theta_5\sin\theta_6 \end{array} & \begin{array}{l} -\sin\theta_3\cos\theta_4\sin\theta_5 \\ +\cos\theta_3\cos\theta_5 \end{array} & \begin{array}{l} d_4\cos\theta_3 \\ +a_3\sin\theta_3 \end{array} \\ -\sin\theta_4\cos\theta_5\cos\theta_6 - \cos\theta_4\sin\theta_6 & \sin\theta_4\cos\theta_5\sin\theta_6 - \cos\theta_4\cos\theta_6 & \sin\theta_4\sin\theta_5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{cases} J_{13} = (-d_4\sin\theta_3 + a_3\cos\theta_3)[\sin\theta_3(\cos\theta_4\cos\theta_5\cos\theta_6 - \sin\theta_4\sin\theta_6) + \cos\theta_3\sin\theta_5\cos\theta_6] \\ \qquad - (d_4\cos\theta_3 + a_3\sin\theta_3)[\cos\theta_3(\cos\theta_4\cos\theta_5\cos\theta_6 - \sin\theta_4\sin\theta_6) - \sin\theta_3\sin\theta_5\cos\theta_6] \\ J_{23} = (-d_4\sin\theta_3 + a_3\cos\theta_3)[-\sin\theta_3(\cos\theta_4\cos\theta_5\sin\theta_6 + \sin\theta_4\cos\theta_6) - \cos\theta_3\sin\theta_5\sin\theta_6] \\ \qquad - (d_4\cos\theta_3 + a_3\sin\theta_3)[-\cos\theta_3(\cos\theta_4\cos\theta_5\sin\theta_6 + \sin\theta_4\cos\theta_6) + \sin\theta_3\sin\theta_5\sin\theta_6] \\ J_{33} = (-d_4\sin\theta_3 + a_3\cos\theta_3)[-\sin\theta_3\cos\theta_4\sin\theta_5 + \cos\theta_3\cos\theta_5] \\ \qquad - (d_4\cos\theta_3 + a_3\sin\theta_3)[-\cos\theta_3\cos\theta_4\sin\theta_5 - \sin\theta_3\cos\theta_5] \\ J_{43} = -\sin\theta_4\cos\theta_5\cos\theta_6 - \cos\theta_4\sin\theta_6 \\ J_{53} = \sin\theta_4\cos\theta_5\sin\theta_6 - \cos\theta_4\cos\theta_6 \\ J_{63} = \sin\theta_4\sin\theta_5 \end{cases}$$

2.2 PUMA560机器人的雅可比矩阵

$$^{1}T_6 = T_2T_3T_4T_5T_6 = \begin{bmatrix} b_{111} & b_{112} & b_{113} & -d_4\sin(\theta_2+\theta_3)+a_2\cos\theta_2+a_3\cos(\theta_2+\theta_3) \\ b_{121} & b_{122} & b_{123} & d_4\cos(\theta_2+\theta_3)+a_2\sin\theta_2+a_3\sin(\theta_2+\theta_3) \\ b_{131} & b_{132} & b_{133} & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$b_{111} = \cos(\theta_2+\theta_3)(\cos\theta_4\cos\theta_5\cos\theta_6 - \sin\theta_4\sin\theta_6) - \sin(\theta_2+\theta_3)\sin\theta_5\cos\theta_6$$

$$b_{121} = \sin(\theta_2+\theta_3)(\cos\theta_4\cos\theta_5\cos\theta_6 - \sin\theta_4\sin\theta_6) + \cos(\theta_2+\theta_3)\sin\theta_5\cos\theta_6$$

$$b_{131} = -\sin\theta_4\cos\theta_5\cos\theta_6 - \cos\theta_4\sin\theta_6$$

$$b_{112} = -\cos(\theta_2+\theta_3)(\cos\theta_4\cos\theta_5\sin\theta_6 + \sin\theta_4\cos\theta_6) + \sin(\theta_2+\theta_3)\sin\theta_5\sin\theta_6$$

$$b_{122} = -\sin(\theta_2+\theta_3)(\cos\theta_4\cos\theta_5\sin\theta_6 + \sin\theta_4\cos\theta_6) - \cos(\theta_2+\theta_3)\sin\theta_5\sin\theta_6$$

$$b_{132} = \sin\theta_4\cos\theta_5\sin\theta_6 - \cos\theta_4\cos\theta_6$$

$$b_{113} = -\cos(\theta_2+\theta_3)\cos\theta_4\sin\theta_5 - \sin(\theta_2+\theta_3)\cos\theta_5$$

$$b_{123} = -\sin(\theta_2+\theta_3)\cos\theta_4\sin\theta_5 + \cos(\theta_2+\theta_3)\cos\theta_5$$

$$b_{133} = \sin\theta_4\sin\theta_5$$

2.2 PUMA560机器人的雅可比矩阵

$$
\begin{cases}
J_{12} = [-d_4 \sin(\theta_2+\theta_3) + a_2 \cos\theta_2 + a_3 \cos(\theta_2+\theta_3)]b_{121} \\
\qquad -[d_4 \cos(\theta_2+\theta_3) + a_2 \sin\theta_2 + a_3 \sin(\theta_2+\theta_3)]b_{111} \\
J_{22} = [-d_4 \sin(\theta_2+\theta_3) + a_2 \cos\theta_2 + a_3 \cos(\theta_2+\theta_3)]b_{122} \\
\qquad -[d_4 \cos(\theta_2+\theta_3) + a_2 \sin\theta_2 + a_3 \sin(\theta_2+\theta_3)]b_{112} \\
J_{32} = [-d_4 \sin(\theta_2+\theta_3) + a_2 \cos\theta_2 + a_3 \cos(\theta_2+\theta_3)]b_{123} \\
\qquad -[d_4 \cos(\theta_2+\theta_3) + a_2 \sin\theta_2 + a_3 \sin(\theta_2+\theta_3)]b_{113} \\
J_{42} = b_{131} \\
J_{52} = b_{132} \\
J_{62} = b_{133}
\end{cases}
$$

2.2 PUMA560机器人的雅可比矩阵

$$^{0}T_6 = T_1T_2T_3T_4T_5T_6$$

$$= \begin{bmatrix} \cos\theta_1 & 0 & -\sin\theta_1 & 0 \\ \sin\theta_1 & 0 & \cos\theta_1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} b_{111} & b_{112} & b_{113} & -d_4\sin(\theta_2+\theta_3)+a_2\cos\theta_2+a_3\cos(\theta_2+\theta_3) \\ b_{121} & b_{122} & b_{123} & d_4\cos(\theta_2+\theta_3)+a_2\sin\theta_2+a_3\sin(\theta_2+\theta_3) \\ b_{131} & b_{132} & b_{133} & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

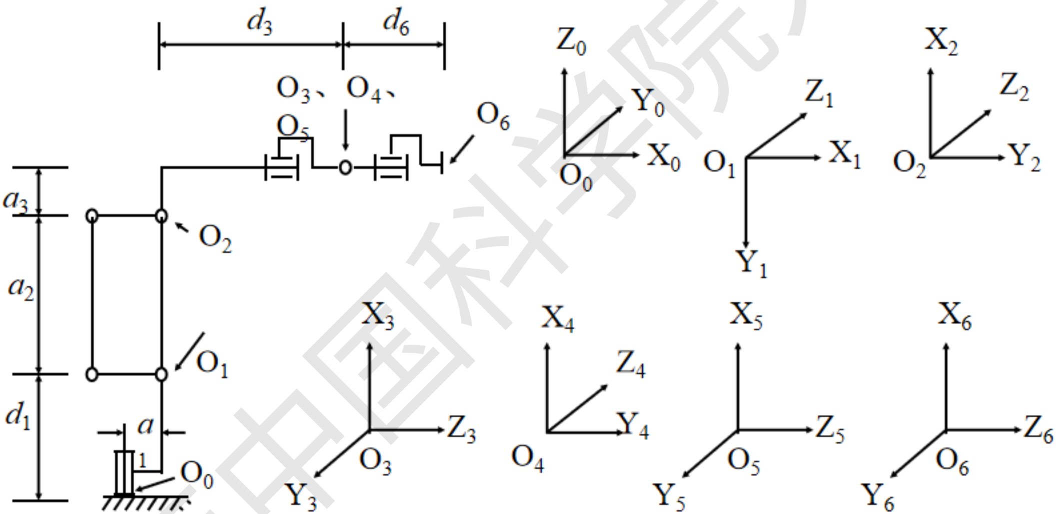
$$= \begin{bmatrix} b_{111}\cos\theta_1 - b_{131}\sin\theta_1 & b_{112}\cos\theta_1 - b_{132}\sin\theta_1 & b_{113}\cos\theta_1 - b_{133}\sin\theta_1 & b_{114}\cos\theta_1 - d_2\sin\theta_1 \\ b_{111}\sin\theta_1 + b_{131}\cos\theta_1 & b_{112}\sin\theta_1 + b_{132}\cos\theta_1 & b_{113}\sin\theta_1 + b_{133}\cos\theta_1 & b_{114}\sin\theta_1 + d_2\cos\theta_1 \\ -b_{121} & -b_{122} & -b_{123} & -b_{124} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{cases} J_{11} = (b_{114}\cos\theta_1 - d_2\sin\theta_1)(b_{111}\sin\theta_1 + b_{131}\cos\theta_1) \\ \qquad -(b_{114}\sin\theta_1 + d_2\cos\theta_1)(b_{111}\cos\theta_1 - b_{131}\sin\theta_1) \\ J_{21} = (b_{114}\cos\theta_1 - d_2\sin\theta_1)(b_{112}\sin\theta_1 + b_{132}\cos\theta_1) \\ \qquad -(b_{114}\sin\theta_1 + d_2\cos\theta_1)(b_{112}\cos\theta_1 - b_{132}\sin\theta_1) \\ J_{31} = (b_{114}\cos\theta_1 - d_2\sin\theta_1)(b_{113}\sin\theta_1 + b_{133}\cos\theta_1) \\ \qquad -(b_{114}\sin\theta_1 + d_2\cos\theta_1)(b_{113}\cos\theta_1 - b_{133}\sin\theta_1) \\ J_{41} = -b_{121} \\ J_{51} = -b_{122} \\ J_{61} = -b_{123} \end{cases}$$

$$\begin{cases} b_{114} = -d_4\sin(\theta_2+\theta_3)+a_2\cos\theta_2+a_3\cos(\theta_2+\theta_3) \\ b_{124} = d_4\cos(\theta_2+\theta_3)+a_2\sin\theta_2+a_3\sin(\theta_2+\theta_3) \end{cases}$$

2.3 K10机器人的雅可比矩阵

YASKAWA机器人的坐标系



YASKAWA K10工业机器人结构与坐标系

2.3 K10机器人的雅可比矩阵

YASKAWA机器人的各个连杆的连杆参数

参数	关节1	关节2	关节3	关节 4	关节5	关节6
a_i	200	600	115	0	0	0
d_i	585	0	770	0	0	100
α_i	-90°	0	-90°	90°	-90°	0
θ_i	θ_1	$-90°+\theta_2$	θ_3	θ_4	θ_5	θ_6

2.3 K10机器人的雅可比矩阵

✓ 各个连杆的变换矩阵

$$T_1=\begin{bmatrix}\cos\theta_1 & 0 & -\sin\theta_1 & a_1\cos\theta_1\\ \sin\theta_1 & 0 & \cos\theta_1 & a_1\sin\theta_1\\ 0 & -1 & 0 & d_1\\ 0 & 0 & 0 & 1\end{bmatrix}\qquad T_2=\begin{bmatrix}\sin\theta_2 & \cos\theta_2 & 0 & a_2\sin\theta_2\\ -\cos\theta_2 & \sin\theta_2 & 0 & -a_2\cos\theta_2\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 1\end{bmatrix}$$

$$T_3=\begin{bmatrix}\cos\theta_3 & 0 & -\sin\theta_3 & a_3\cos\theta_3-d_3\sin\theta_3\\ \sin\theta_3 & 0 & \cos\theta_3 & a_3\sin\theta_3+d_3\cos\theta_3\\ 0 & -1 & 0 & 0\\ 0 & 0 & 0 & 1\end{bmatrix}$$

$$T_4=\begin{bmatrix}\cos\theta_4 & 0 & \sin\theta_4 & 0\\ \sin\theta_4 & 0 & -\cos\theta_4 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1\end{bmatrix}\quad T_5=\begin{bmatrix}\cos\theta_5 & 0 & -\sin\theta_5 & 0\\ \sin\theta_5 & 0 & \cos\theta_5 & 0\\ 0 & -1 & 0 & 0\\ 0 & 0 & 0 & 1\end{bmatrix}\quad T_6=\begin{bmatrix}\cos\theta_6 & -\sin\theta_6 & 0 & 0\\ \sin\theta_6 & \cos\theta_6 & 0 & 0\\ 0 & 0 & 1 & d_6\\ 0 & 0 & 0 & 1\end{bmatrix}$$

2.3 K10机器人的雅可比矩阵

✓ 连杆到机器人末端的变换矩阵与Jacobian矩阵

$$J_i = [(p\times n)_z \quad (p\times o)_z \quad (p\times a)_z \quad n_z \quad o_z \quad a_z]^T$$

$$^5T_6 = T_5T_6 = \begin{bmatrix} \cos\theta_5 & 0 & -\sin\theta_5 & 0 \\ \sin\theta_5 & 0 & \cos\theta_5 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} \cos\theta_6 & -\sin\theta_6 & 0 & 0 \\ \sin\theta_6 & \cos\theta_6 & 0 & 0 \\ 0 & 0 & 1 & d_6 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos\theta_5\cos\theta_6 & -\cos\theta_5\sin\theta_6 & -\sin\theta_5 & -\sin\theta_5 d_6 \\ \sin\theta_5\cos\theta_6 & -\sin\theta_5\sin\theta_6 & \cos\theta_5 & \cos\theta_5 d_6 \\ -\sin\theta_6 & -\cos\theta_6 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$^4T_6 = T_4\,{}^5T_6 = \begin{bmatrix} \cos\theta_4\cos\theta_5\cos\theta_6 - \sin\theta_4\sin\theta_6 & -\cos\theta_4\cos\theta_5\sin\theta_6 - \sin\theta_4\cos\theta_6 & -\cos\theta_4\sin\theta_5 & -\cos\theta_4\sin\theta_5 d_6 \\ \sin\theta_4\cos\theta_5\cos\theta_6 + \cos\theta_4\sin\theta_6 & -\sin\theta_4\cos\theta_5\sin\theta_6 + \cos\theta_4\cos\theta_6 & -\sin\theta_4\sin\theta_5 & -\sin\theta_4\sin\theta_5 d_6 \\ \sin\theta_5\cos\theta_6 & -\sin\theta_5\sin\theta_6 & \cos\theta_5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ p_x & p_y & p_z \\ n_x & n_y & n_z \end{vmatrix} = (p_y n_z - p_z n_y)\vec{i} - (p_x n_z - p_z n_x)\vec{j} + (p_x n_y - p_y n_x)\vec{k}$$

$$J_6 = [0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1]^T$$

$$J_5 = [-\cos\theta_6 d_6 \quad \sin\theta_6 d_6 \quad 0 \quad -\sin\theta_6 \quad -\cos\theta_6 \quad 0]^T$$

$$J_4 = [-\sin\theta_5\sin\theta_6 d_6 \quad -\sin\theta_5\cos\theta_6 d_6 \quad 0 \quad \sin\theta_5\cos\theta_6 \quad -\sin\theta_5\sin\theta_6 \quad \cos\theta_5]^T$$

3 机器人的静力变换

- 静力和力矩的表示：F广义力

$$F=\begin{bmatrix} f_x & f_y & f_z & m_x & m_y & m_z \end{bmatrix}^T$$

静力， 力矩

- 不同坐标系间静力和力矩的变换

对于基坐标系{B},有：

虚功$\delta W = F^T D$, D为虚拟位移

$$F=\begin{bmatrix} f_x & f_y & f_z & m_x & m_y & m_z \end{bmatrix}^T$$

D为虚拟位移, $D=\begin{bmatrix} d_x & d_y & d_z & \delta_x & \delta_y & \delta_z \end{bmatrix}^T$

对于坐标系{C},有：

$$\delta W = F^T D = {}^cF^T\,{}^cD$$

$${}^cF=\begin{bmatrix} {}^cf_x & {}^cf_y & {}^cf_z & {}^cm_x & {}^cm_y & {}^cm_z \end{bmatrix}^T$$

C下的虚拟位移 ${}^cD=\begin{bmatrix} {}^cd_x & {}^cd_y & {}^cd_z & {}^c\delta_x & {}^c\delta_y & {}^c\delta_z \end{bmatrix}^T$

根据等价微分变换,可以得到等价的力和力矩

3 机器人的静力变换

$$
\begin{bmatrix} {}^T d_x \\ {}^T d_y \\ {}^T d_z \\ {}^T \delta_x \\ {}^T \delta_y \\ {}^T \delta_z \end{bmatrix} = \begin{bmatrix} n_x & n_y & n_z & (p\times n)_x & (p\times n)_y & (p\times n)_z \\ o_x & o_y & o_z & (p\times o)_x & (p\times o)_y & (p\times o)_z \\ a_x & a_y & a_z & (p\times a)_x & (p\times a)_y & (p\times a)_z \\ 0 & 0 & 0 & n_x & n_y & n_z \\ 0 & 0 & 0 & o_x & o_y & o_z \\ 0 & 0 & 0 & a_x & a_y & a_z \end{bmatrix} \begin{bmatrix} d_x \\ d_y \\ d_z \\ \delta_x \\ \delta_y \\ \delta_z \end{bmatrix} \Rightarrow {}^c D = \Gamma D
$$

$$
\begin{aligned}
& F^T D = {}^c F^T \, {}^c D \\
\Rightarrow\ & F^T D = {}^c F^T \, \Gamma D \\
\Rightarrow\ & F^T = {}^c F^T \, \Gamma \\
\Rightarrow\ & F = \Gamma^T \, {}^c F \\
\Rightarrow\ & {}^c F = (\Gamma^T)^{-1} F
\end{aligned}
$$

$$
\begin{bmatrix} {}^c f_x \\ {}^c f_y \\ {}^c f_z \\ {}^c m_x \\ {}^c m_y \\ {}^c m_z \end{bmatrix} = \begin{bmatrix} n_x & o_x & a_x & 0 & 0 & 0 \\ n_y & o_y & a_y & 0 & 0 & 0 \\ n_z & o_z & a_z & 0 & 0 & 0 \\ (p\times n)_x & (p\times o)_x & (p\times a)_x & n_x & o_x & a_x \\ (p\times n)_y & (p\times o)_y & (p\times a)_y & n_y & o_y & a_y \\ (p\times n)_z & (p\times o)_z & (p\times a)_z & n_z & o_z & a_z \end{bmatrix}^{-1} \begin{bmatrix} f_x \\ f_y \\ f_z \\ m_x \\ m_y \\ m_z \end{bmatrix}
$$

3 机器人的静力变换

$$\begin{bmatrix} {}^{c}f_x \\ {}^{c}f_y \\ {}^{c}f_z \\ {}^{c}m_x \\ {}^{c}m_y \\ {}^{c}m_z \end{bmatrix} = \begin{bmatrix} n_x & n_y & n_z & 0 & 0 & 0 \\ o_x & o_y & o_z & 0 & 0 & 0 \\ a_x & a_y & a_z & 0 & 0 & 0 \\ (p\times n)_x & (p\times n)_y & (p\times n)_z & n_x & n_y & n_z \\ (p\times o)_x & (p\times o)_y & (p\times o)_z & o_x & o_y & o_z \\ (p\times a)_x & (p\times a)_y & (p\times a)_z & a_x & a_y & a_z \end{bmatrix} \begin{bmatrix} f_x \\ f_y \\ f_z \\ m_x \\ m_y \\ m_z \end{bmatrix}$$

$\Downarrow$ 调整向量中元素顺序

$$\begin{bmatrix} {}^{c}m_x \\ {}^{c}m_y \\ {}^{c}m_z \\ {}^{c}f_x \\ {}^{c}f_y \\ {}^{c}f_z \end{bmatrix} = \begin{bmatrix} n_x & n_y & n_z & (p\times n)_x & (p\times n)_y & (p\times n)_z \\ o_x & o_y & o_z & (p\times o)_x & (p\times o)_y & (p\times o)_z \\ a_x & a_y & a_z & (p\times a)_x & (p\times a)_y & (p\times a)_z \\ 0 & 0 & 0 & n_x & n_y & n_z \\ 0 & 0 & 0 & o_x & o_y & o_z \\ 0 & 0 & 0 & a_x & a_y & a_z \end{bmatrix} \begin{bmatrix} m_x \\ m_y \\ m_z \\ f_x \\ f_y \\ f_z \end{bmatrix} \Rightarrow \begin{bmatrix} {}^{c}m_x \\ {}^{c}m_y \\ {}^{c}m_z \\ {}^{c}f_x \\ {}^{c}f_y \\ {}^{c}f_z \end{bmatrix} = \Gamma \begin{bmatrix} m_x \\ m_y \\ m_z \\ f_x \\ f_y \\ f_z \end{bmatrix}$$

机器人的稳态负荷

- 关节力矩的确定

$$\delta W = {}^{T_6}F^T \ {}^{T_6}D \qquad {}^{T_6}D = JQ$$

Q为关节增量

$$\delta W = \tau^T Q$$

${}^{T_6}D$为末端位移增量

$$\tau^T = {}^{T_6}F^T \ J$$

转置

$$\tau = J^T \ {}^{T_6}F$$

串联机器人末端力/力矩与关节力/力矩间映射

其中τ为关节的力（平移关节）或力矩（旋转关节）

末端坐标系下 ${}^{n}\tau = {}^{n}J^T \ {}^{n}F$ 　　基端坐标系下 ${}^{0}\tau = {}^{0}J^T \ {}^{0}F$

速度求逆力转置 $\dot{q} = J^{-1}\dot{x} \qquad \tau = J^T \ {}^{T_6}F$

机器人的稳态负荷

➢ 负荷质量的确定：关节力矩的应用

机器人移动未知负荷时，可由关节误差力矩求此负荷的质量。步骤为：

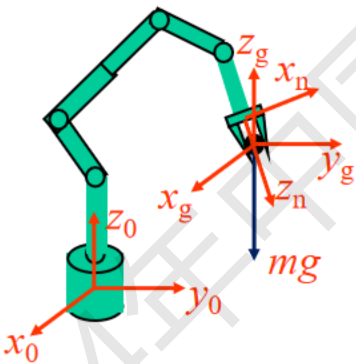
- ✓ 设定速度增益，使之在最大负荷下也不产生欠阻尼响应
- ✓ 机器人以恒速提升该负荷，计算各个关节的静态误差力矩与力τ_{m}
- ✓ 根据不同坐标系之间的静力转换，求取${}^{G}F$对${}^{T_6}F$的微分变换，进而求得T_6上对1千克负荷的作用力。
- ✓ 计算各个关节的等效关节力τ_1
- ✓ 计算负荷质量

$$\boldsymbol{\tau} = {}^{n}J^{T}\ {}^{n}F \qquad {}^{g}F = \begin{bmatrix} 0 & 0 & -mg & 0 & 0 & 0 \end{bmatrix}^{T}$$

若已知 ${}^{n}T_{g}$ 则按微分变换可知 ${}^{g}F = \Gamma\ {}^{n}F$

$$\boldsymbol{\tau} = {}^{n}J^{T}\ \Gamma^{-1}\ {}^{g}F = A\begin{bmatrix} 0 & 0 & -mg & 0 & 0 & 0 \end{bmatrix}^{T}$$

$$A = {}^{n}J^{T}\ \Gamma^{-1} = \begin{bmatrix} \boldsymbol{a}_1 & \boldsymbol{a}_2 & \boldsymbol{a}_3 & \boldsymbol{a}_4 & \boldsymbol{a}_5 & \boldsymbol{a}_6 \end{bmatrix}$$



假设计算τ_{m}与τ_1时的J相同

$$\boldsymbol{\tau}_m = a_3 \cdot (-mg) \qquad \boldsymbol{\tau}_1 = a_3 \cdot (-g) \qquad m^2 = \frac{\tau_m^T \tau_m}{\tau_1^T \tau_1}$$

THANK YOU